MATH 420-01

DISCUSS MIDTERM

MIDTERM COVERS Sections 1.1-1.6, 2.1-2.5, and Three Ways to Multiply Matrices.

Note that I will not test on 1.7, the section with maximal sets and Zorn's lemma. The third week of October-ish

W – 18	Th – 19	F – 20	Sa	Su	M – 23	Tu – 24	W – 25	Th – 26	F – 27
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One side of one 8.5x11 sheet of notes.

Our "day off" will be November 18?

2.4 Invertibility and Isomorphisms

Recall:

$I_V : V \to V$ is the linear transformation such that $I_V(v) = v,\ \forall\, v \in V$.

Example:

Find $[I_V]_\alpha^\beta$ when $\alpha = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ and $\beta = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

apply transformation

Example:

Find $[I_V]_\alpha^\beta$ when $\alpha = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ and $\beta = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

$$\begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 1 \\ 2 & 0 & 1 & 1 & -1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

we need lin combs of β vectors

In[ ]:= $\text{RowReduce}\left[\begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 1 \\ 2 & 0 & 1 & 1 & -1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}\right]$

Out[ ]= $\begin{pmatrix} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{pmatrix}$

We could have solved three systems:

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 2 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 & 1 \\ 2 & 0 & 1 & -1 \\ 1 & 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 & 1 \\ 2 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$$[I_V]_\alpha^\beta = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

$$[I_V]_\beta^\beta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Definition:

Let V and W be vector spaces over a field $\mathbb{F}$. Let $T : V \to W$ be a linear transformation. A transformation $S : W \to V$ is called the *inverse* of T if $S\,T = I_V$ and $T\,S = I_W$. If T has an inverse, it is said to be *invertible*, and the inverse is denoted by T^{-1}.

Invertible and nonsingular are synonyms

Not Invertible and singular are synonyms.

Something is invertible or it is NOT.

Theorem:

Let V and W be vector spaces over a field $\mathbb{F}$. Let $T : V \to W$ be an invertible linear transformation. Then

(i) $T^{-1} : W \to V$ is a linear transformation.

(ii) $\left(T^{-1}\right)^{-1} = T$.

(iii) If V is finite dimensional, then $\dim(W) = \dim(V)$.

(iv) If $S : W \to V$ is an invertible linear transformation, then $(S\,T)^{-1} = T^{-1}\,S^{-1}$.

(v) T is one-to-one and onto.

PROOF:

(i) Want to show $T^{-1}(ax + y) = a\,T^{-1}(x) + T^{-1}(y)$ where $a \in \mathbb{F}$ and $x, y \in W$

Notice $a\,T^{-1}(x) + T^{-1}(y) = T^{-1}\,T\left(a\,T^{-1}(x) + T^{-1}(y)\right) = T^{-1}\left(a\,T\,T^{-1}(x) + T\,T^{-1}(y)\right) = T^{-1}(ax + y)$, thus T^{-1} is linear.

(ii) Since T is the thing you compose with T^{-1} to get the identity, it is the inverse's inverse.

(iii) Assume V is finite dimensional. Let $\{v_1, \ldots v_n\}$ be a basis for V. Then $\{T(v_1), \ldots, T(v_n)\}$ is a spanning set for the Range of T and moreover, $\left\{T^{-1}\,T(v_1), \ldots, T^{-1}\,T(v_n)\right\} = \{v_1, \ldots v_n\}$, a basis for V.

From this we know that T^{-1} maps the range of T onto V, however we are going take a different approach:

$\dim W = \operatorname{rank}\left(T\,T^{-1}\right) \le \operatorname{rank}(T) \le \operatorname{rank}(T) + \operatorname{nullity}(T) = \dim V$

$\dim V = \operatorname{rank}\left(T^{-1}\,T\right) \le \operatorname{rank}\left(T^{-1}\right) \le \operatorname{rank}\left(T^{-1}\right) + \operatorname{nullity}\left(T^{-1}\right) = \dim W$

Thus everything is equal throughout, establishing (iii) and (v) since $\operatorname{rank}(T) = \dim(W)$ and $\operatorname{nullity}(T) = 0$.